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Joint Wind–Photovoltaic Power Forecasting Based on MultiTask Learning With Partial Site Observations

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ABSTRACT

Accurate and efficient forecasts can help address uncertainties posed by distributed renewable power generation (DRPG). Forecasting for DRPG sites requires considering several critical factors, such as large quantity, strong spatio-temporal correlation and scarce historical measurements. To this end, a framework for jointly forecasting the outputs of multiple wind and photovoltaic (PV) power generation sites is proposed, particularly for cases where not all sites have sufficient historical measurements. Firstly, the Kriging method is used to interpolate the data for sites without any measurement and the measure correlate–predict (MCP) method to derive long-term data for sites with only short-term measurements. Then three joint forecasting architectures are applied and compared to forecast outputs of wind and PV sites so as to improve the modelling efficiency by aggregating different forecasting tasks into one multi-task learning model and improve forecasting accuracy by exploiting spatial correlation among different sites. The framework is tested using data from 15 wind sites and 15 PV sites. Results demonstrate that the Kriging and MCP methods are effective for estimating data for DRPG sites without enough history measurements to build forecasting models. The multi-task regression model based on the least absolute shrinkage and selection operator achieves the highest accuracy among the three studied architectures.

1 | Introduction

Distributed renewable power generation (DRPG) is an essential supplement to the centralised form since it is more flexible to develop and control and could reduce the losses due to transmission and distribution of electricity [1]. Among the main types of distributed renewable sources, distributed wind and photovoltaic (PV) power have experienced rapid growth in installed capacity worldwide over the past decades [2–4].

The distributed wind and PV power generations (DWPG) are intrinsically uncertain and fluctuate because they are highly dependent on weather states. The high penetration of DWPG in distribution networks can lead to various operational challenges, such as voltage instability [5] and reverse power flow [6]. The key to addressing these challenges is to accurately and efficiently forecast the DWPG and to make optimal decisions in advance based on the forecasts [7].

Most research works on renewable energy forecasting are conducted for centralised wind farms and PV sites to support decision-making at the transmission level [8]. Especially in recent years, the spatial correlation is frequently adopted to incorporate the valuable information from neighbouring reference sites to improve the forecasting accuracy of target sites. A correlation-constrained and sparsity-controlled vector autoregressive model was proposed in [9] for spatio-temporal wind power forecasting to control the selection of neighbouring reference sites and their history data. Hu et al. [10] developed a convolution-based spatial-temporal wind power predictor for very short-term forecasting, where a convolution operation is used to capture the spatial and temporal correlations among multiple wind farms. The

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graph deep neural network is another important forecasting tool to learn spatio-temporal features from weather and power generation data [11, 12].

Many proven forecasting techniques for centralised wind and PV power are also applicable to DRPG. However, the DWPG sites have their own distinctive characteristics when dealing with forecasting issues, as follows.

1. In comparison with the centralised wind and PV power stations, the quantity of DWPG units is large while the individual capacity is generally small, and the different distributed units in a small region could belong to different and diverse owners, e.g., a small wind turbine operated by a hospital or a university, or rooftop solar panels owned by a householder. In this case, it is inefficient, and the owners are less motivated to build a specific forecasting model for each individual unit.
2. Simultaneously forecasting distributed wind and PV power by leveraging their potentially correlated information presents a promising approach to enhancing accuracy, particularly given their co-location within a limited area and exposure to similar weather patterns [13].
3. Large-scale wind and PV farms benefit from numerical weather prediction (NWP) data and on-site measurements. In contrast, many DWPG units are newly installed with little or no data, and most are behind-the-meter [14], making them unmonitored by distribution system operators. This lack of data poses a major challenge for forecasting, which relies heavily on extensive training data.

Although distributed renewable generation is advancing rapidly, related forecasting research is still limited. Existing studies on distributed wind or PV forecasting do not fully address the critical challenges mentioned above. Many joint forecasting and multi-task learning methods have been proposed for source-load-price forecasting [15, 16] and multi-energy load forecasting [17, 18], but they are generally not for distributed renewable energy, and no data problem is reported. A temporally distributed model named gated recurrent unit pool was proposed in [19] for distributed PV power forecasting, but it is an autocorrelation-based model, and the spatial correlation is not considered. The spatio-temporal correlation between PV power plants is adopted in [20] for ultra-short-term PV power forecasting by using neighbouring plants to forecast the power of target plants. But the model is for only one target plant at a time, which is inefficient for modelling all plants. In [21], graph-convolutional long short-term memory and graph-convolutional transformer models were developed for multi-site PV power forecasting, using production data from 304 real PV systems. However, the truth that data are not always available for every PV system in distribution networks is neglected, and the forecasting is for a single energy form. In [22], a short-term regional distributed PV power forecasting method was proposed based on sub-region division and spatio-temporal correlations between sub-regions. Similarly, regional distributed PV power was forecasted in [23] by using a dynamic directed graph convolution network. But they did not consider the issue of data scarcity. A heterogeneous spatiotemporal graph convolution network is proposed in [24] for wind–PV power collaborative prediction, but it is designed for large centralised power generation sites and also relies on sufficient NWP data and power measurements. Distributed wind power forecasting was studied in [25] by using high-precision NWP. But it is always difficult to obtain NWP for DRPG. In [26] and [27], the distributed PV power forecasting is focused on the data-scarce scenarios by transferring forecasting-related knowledge from sites of data-rich regions to the data-scarce region. But its application scope may be limited, and it is only for one renewable resource.

In summary, existing studies exhibit one or more of the following shortcomings:

1. Forecasting is limited to a single type of renewable resource and only one target generation site or unit at a time. As a result, the modelling lacks efficiency, and interactions between different renewable energy resources are overlooked.
2. Spatial correlations among neighbouring renewable power generation units are not accounted for to improve forecasting accuracy.
3. The issues of data scarcity for training forecasting models are not addressed, potentially rendering the proposed models impractical for real-world applications.

To fully consider the features of distributed wind and PV power generation so as to build a multi-energy-form and multi-site forecasting model using correlated information as input, a framework for joint forecasting of wind–PV power generation is proposed. In this framework, different data interpolation methods are applied for the distributed generation locations without complete data, considering different cases, i.e., the locations to be interpolated have accumulated either no data or only very short-term data. Then, different types and different architectures of joint forecasting models are developed and compared to find appropriate configurations. The framework is tested on the data from 15 distributed wind power generation sites and 15 distributed PV power generation sites over the same region.

The primary objective of this work is not to introduce a new algorithm or forecasting model. Rather, a novel framework for joint wind–PV power forecasting is proposed, particularly suited for scenarios with generally incomplete measurement data. Additionally, a comprehensive comparative analysis of the effectiveness and adaptability of various machine learning model architectures in this context is done in order to give insights for developing more advanced joint forecasting models.

2 | Spatial Interpolation Methods for Measurements Reconstruction

In this section, two methods are introduced to interpolate data for distributed wind and PV sites with incomplete measurements. The Kriging method is employed to estimate values at locations lacking any data, based on available data from neighbouring sites. The measure–correlate–predict (MCP) method is used to generate long-term data at sites where only short-term measurements are available.

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2.1 | Kriging Method for Locations Without Measurements

Spatial interpolation exploits spatial autocorrelation in order to predict an unknown value at some location from known values at nearby locations. Kriging is known as a best linear unbiased predictor for optimal spatial interpolation used in geostatistics, based on a stochastic model of spatial variation [28]. It makes best use of what can be inferred about the spatial structure in the spatial surface to be interpolated from an analysis of data at the control points or sampling points [29]. It was developed in the 1960s by Georges Matheron and was named after Dani Krige.

Several variants of the Kriging method exist, such as Ordinary Kriging, Simple Kriging, and Disjunctive Kriging, differentiated by whether the mean is known and whether the method is linear or non-linear. Ordinary Kriging assumes an unknown mean and local stationarity of the process. Due to its robustness and widespread application, it is adopted for interpolation in this study.

Consider that a random variable, Z (e.g., distributed wind power generation or wind speed), has been measured at sampling locations, $x_i, i = 1, 2, \ldots, n$. The objective is to use the measurements of Z at sampling locations to estimate the measurement value of Z at an unsampled location x_0, which can be expressed as a weighted sum of the Z values of sampling locations.

$$\begin{aligned}\hat{Z}(x_0) &= w_1 Z(x_1) + w_2 Z(x_2) + \cdots + w_n Z(x_n) \\ &= \sum_{i=1}^{n} w_i Z(x_i) = \boldsymbol{w}^{\mathrm{T}} \boldsymbol{Z}\end{aligned} \tag{1}$$

where w_1, w_2, …, w_n are weights of sampled values used to estimate the interpolation value, and n represents the number of sampling locations within the local neighbourhood of x_0. To ensure that the estimate is unbiased, the weights are made to sum to one [30].

$$\sum_{i=1}^{n} w_i = 1 \tag{2}$$

The expected estimation error is $E[\hat{Z}(x_0) - Z(x_0)] = 0$, and the variance is

$$\begin{aligned}\mathrm{var}\left[\hat{Z}(x_0)\right] &= E\left[\left\{\hat{Z}(x_0) - Z(x_0)\right\}^2\right] \\ &= 2\sum_{i=1}^{n} w_i \gamma(x_i, x_0) - \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \gamma(x_i, x_j)\end{aligned} \tag{3}$$

where $\gamma(x_i, x_0)$ is the semivariance of Z between sampling location x_i and target unsampled location x_0, and $\gamma(x_i, x_j)$ is the semivariance between the sampling locations x_i and x_j. The semivariance can be derived from a variogram model.

The estimation error is minimised and thus the weights will be obtained if the following system of linear equations is solved for unknown weights and an introduced Lagrangian multiplier λ:

$$\begin{cases} w_1\gamma(x_1, x_1) + w_2\gamma(x_1, x_2) + \cdots + w_n\gamma(x_1, x_n) + \lambda = \gamma(x_1, x_0) \\ \vdots \\ w_1\gamma(x_n, x_1) + w_2\gamma(x_n, x_2) + \cdots + w_n\gamma(x_n, x_n) + \lambda = \gamma(x_n, x_0) \\ w_1 + w_2 + \cdots w_n + 0 = 1 \end{cases} \tag{4}$$

The above equations could be written in matrix form

$$\begin{bmatrix} \gamma(x_1, x_1) & \gamma(x_1, x_2) & \ldots & \gamma(x_1, x_n) & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \gamma(x_n, x_1) & \gamma(x_n, x_2) & \ldots & \gamma(x_n, x_n) & 1 \\ 1 & 1 & \ldots & 1 & 0 \end{bmatrix} \times \begin{bmatrix} w_1 \\ \vdots \\ w_n \\ \lambda \end{bmatrix} = \begin{bmatrix} \gamma(x_1, x_0) \\ \vdots \\ \gamma(x_n, x_0) \\ 1 \end{bmatrix} \tag{5}$$

which is represented in a compact form

$$\boldsymbol{\Gamma} w = \boldsymbol{\gamma} \tag{6}$$

The weights are solved by

$$\mathbf{w} = \boldsymbol{\Gamma}^{-1}\boldsymbol{\gamma} \tag{7}$$

The optimised weights are then used to compute the interpolation values at location x_0 using Equation (1) and based on the measurements from sampling locations.

It should be noted that the Kriging assumes spatial stationarity, which may not hold in heterogeneous terrains or under rapidly changing environmental conditions. It also depends on the choice and fitting of the variogram model, which can be challenging in practice. Moreover, Kriging is sensitive to spatial outliers, which can distort interpolation accuracy. It also requires relatively dense and high-quality spatial data; otherwise, the estimation uncertainty increases significantly.

2.2 | MCP Method for Locations With Partial Short-Term Measurements

The MCP method was originally developed for, and has been widely applied in, wind resource assessment. Its primary goal is to perform long-term hindcasting at a candidate site where only short-term wind data are available. The approach relies on long-term historical time series from nearby reference sites. A key requirement of the method is that a portion of the long-term data from the reference sites must overlap in both time and duration with the short-term data at the target site. This overlap enables the establishment of a statistical correlation between the target and reference sites [31].

The whole procedure of the MCP consists of three stages, as illustrated in Figure 1.

1. Data collection. Short-term synchronous time series are collected from both the reference sites and the target site, while long-term time series are obtained from the reference sites only.
2. Building a correlation model. This forms the core of the MCP method. The key step is to determine the weights of

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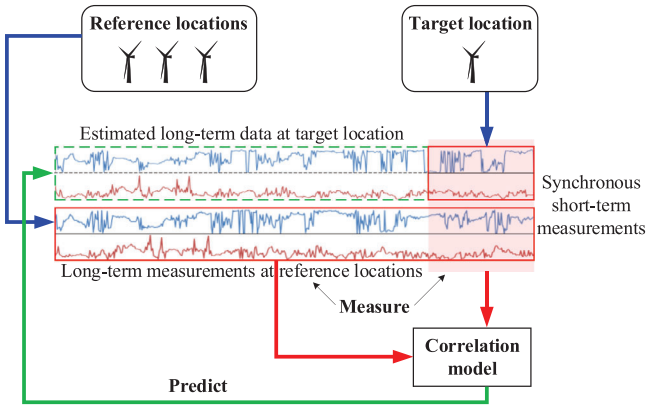


FIGURE 1 | Illustration of the MCP method.

the reference site measurements for estimating data at the target site, based on the short-term synchronous time series. Common approaches for deriving these weights include correlation analysis and ranking using statistical metrics (e.g., Pearson correlation), multivariate linear regression (MLR) models that treat reference site features as regressors, and machine learning models that use inputs from the reference sites [32].

3. Estimating long-term data for the target location. The long-term data at the target site can then be estimated by applying the derived weights or correlation model to the long-term time series from the reference sites.

It should be noted that there are differences between the MCP and the Kriging. The Kriging requires cross-sectional measurements of reference locations at a particular time point and the geographical information (coordinates or distances) of all locations. By contrast, the MCP relies on time series data from both reference locations and the target location, while the geographical information is not necessary.

In this work, the MLR model is applied to model the relationship between reference locations and the target location. In this regard, the MLR model for estimation of variable Z shares the same formulation as the Kriging, i.e., Equation (1).

The weights or the coefficients of the MLR model are obtained by minimising the least squares objective function using the short-term time series data from reference and target locations for training

$$\tilde{\boldsymbol{w}} = \arg\min_{\boldsymbol{w}} \sum_{t=1}^{T_s} \left(Z_t(x_0) - \hat{Z}_t(x_0)\right)^2 = \arg\min_{\boldsymbol{w}} \sum_{t=1}^{T_s} \left(Z_t(x_0) - \boldsymbol{w}^{\mathrm{T}}\boldsymbol{Z}_t\right)^2 \quad (8)$$

where T_s is the total number of time points of the short-term time series data, $Z_t(x_0)$ is the Z value of the target location x_0 at time point t, and $\boldsymbol{Z}_t = (Z(x_1), Z(x_2),\ldots, Z(x_n))^{\mathrm{T}}$ is the vector of cross-sectional measurements at all reference locations at time point t.

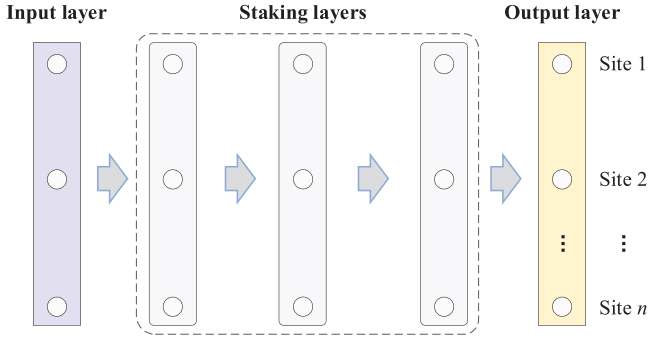


FIGURE 2 | The architecture of traditional multi-input multi-output deep learning.

Based on the optimised weights and the long-term measurements from reference locations, the long-term data for the target location can be estimated

$$\hat{Z}_t(x_0) = \tilde{\boldsymbol{w}}^{\mathrm{T}}\boldsymbol{Z}_t, t = T_s + 1, T_s + 2, \ldots, T_L \quad (9)$$

where T_L is the maximum time index for the long-term time series data to be estimated.

When using MCP, it should be noted that the method assumes a stable linear relationship between the reference and target sites, which may not be valid under non-stationary weather conditions or in areas with complex local influences. Additionally, though MCP is used to derive long-term data of target sites by using short-term data of reference sites, its performance can degrade when the reference sites have low correlation with the target site or when the amount of available data is insufficient to establish a reliable statistical relationship.

3 | Model Architectures for Joint Forecasting

Compared with single-energy or single-site forecasting, joint forecasting of multi-energy-form, multi-site distributed wind and PV power generation offers two main advantages. First, it improves modelling efficiency by eliminating the need to develop separate forecasting models for each energy type and site. Second, it allows the exploitation of spatial correlations between sites and potential interdependencies between energy forms within a given area, as they are often influenced by the same moving weather systems. To explore this, three different model architectures are proposed and compared for joint forecasting of distributed wind–PV power generation.

3.1 | Multi-Input Multi-Output Deep Learning

The multi-input multi-output deep learning (MIMODL) is the most traditional and straightforward machine learning model architecture. The MIMODL has one input layer, one output layer and a stack of intermediate layers, as shown in Figure 2. When it is applied to joint forecasting of distributed wind and PV power generation, the forecasts of different distributed wind and PV sites are given simultaneously by the last single output layer.

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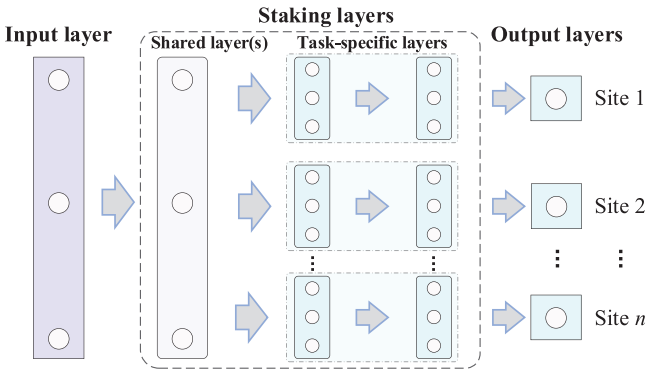


FIGURE 3 | The architecture of multi-task deep learning.

The most notable feature of this architecture is that all output units share the same intermediate layers and corresponding weights. This design enforces the retention of shared information across different energy forms and sites. However, it may also lead to the loss of site-specific or energy-specific features, as the model lacks dedicated pathways to preserve unique characteristics during training.

3.2 | Multi-Task Deep Learning

The multi-task deep learning (MTDL) architecture accounts not only for the shared information among different distributed wind and PV sites but also for the unique and distinct characteristics of power outputs from each site.

In the MTDL architecture, as shown in Figure 3, a stack of lower layers is designed to extract correlated information that is shared by all wind and PV sites. After that, task-specific layers are provided for parallel tasks of forecasting each individual site. Each task-specific layer accepts the outputs from the shared layers and is responsible for extracting information for the forecast of corresponding site. In terms of the forecasting outputs, each task (site) has its own output layer. This is different from the MIMODL architecture, where the outputs of all sites belong to one output layer.

It should be noted that if the number of all task-specific layers becomes zero, the MTDL would be degraded to the MIMODL.

3.3 | Multi-Task LASSO Model

The Least Absolute Shrinkage and Selection Operator (LASSO) is a famous sparsity modelling technique with an l_1-norm regulariser for improving the generalisation ability of forecasting models, by which a large number of weights or parameters of the model are optimised to be zero during training. The LASSO is in essence a feature selection method. However, traditional feature subset selection is a discrete process with high variance and thus can not necessarily improve forecasting accuracy. Comparatively, the LASSO does a kind of continuous subset selection.

The LASSO-based vector autoregression (VAR) model has been used for wind power forecasting in several research works to exploit the spatial correlation of multiple neighbouring wind farms and has shown good performance [9, 33].

The multi-task LASSO (MT-LASSO), based on the traditional LASSO, is specifically used for multiple regression forecasting, which is trained with an l_1 l_2 mixed-norm as a regulariser. It allows fitting multiple regression problems by jointly training the features for different tasks, i.e., forecasting the power outputs of different wind and PV sites in this work. The optimisation objective for the VAR based on MT-LASSO is

$$\min_{\boldsymbol{W}} \frac{1}{2L} \| \boldsymbol{Y} - \boldsymbol{X}\boldsymbol{W} \|_{Fro}^2 + \alpha \| \boldsymbol{W} \|_{21} \tag{10}$$

where $\boldsymbol{X} \in \mathrm{R}^{L \times F}$ is the matrix composed of history measurements of weather variables and power outputs at wind and PV sites, $\boldsymbol{Y} \in \mathrm{R}^{L \times S}$ is the data matrix of power output measurements, $\boldsymbol{W} \in \mathrm{R}^{F \times S}$ is the coefficient matrix to be optimised, α is the regularisation parameter controlling the degree of sparsity of the estimated coefficient matrix $\boldsymbol{W}$, L is the number of training samples, F is the number of features from all sites, S is the total number of wind and PV sites, and the $\|\cdot\|_{Fro}$ indicates the Frobenius norm

$$\| \boldsymbol{A} \|_{\mathrm{Fro}} = \sqrt{\sum_{ij} a_{ij}} \tag{11}$$

where a_{ij} is the element of matrix $\boldsymbol{A}$. The l_1 l_2 mixed-norm is represented as

$$\| \boldsymbol{A} \|_{21} = \sum_{i} \sqrt{\sum_{j} a_{ij}^2} \tag{12}$$

The MT-LASSO objective function could be solved using the coordinate descent algorithm. It is important to note that the joint operation of MT-LASSO differs significantly from that of MTDL. In the MTDL, all tasks share a set of common weights/coefficients via the shared layers, and meanwhile different tasks also have their own distinct coefficients in the task-specific layers. However, in the MT-LASSO, tasks do not share any common coefficients, and each task retains its own distinct group of coefficients for all features. The joint operation of MT-LASSO is only reflected in that the coefficients of different tasks share and contribute to the same loss function.

3.4 | Basic Deep Learning Model

Various deep learning models could be applied to the two deep learning model architectures for joint forecasting introduced above, i.e., the MIMODL and the MTDL. The long short-term memory (LSTM) network is adopted as the basic forecasting model in this paper. The LSTM is one of the most typical recurrent neural network (RNN) models and has been widely used. It can alleviate the vanishing gradient problem present in traditional RNNs. More importantly, it excels in capturing long-range dependencies in sequential data due to its memory cells and gating mechanisms, and it is flexible with variable input lengths. These features make LSTM models well-suited for renewable energy forecasting.

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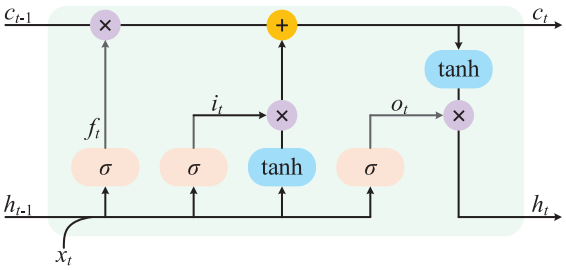


FIGURE 4 | The structure of an LSTM cell.

An LSTM model typically consists of a sequence of LSTM cells that maintain information over time. Each LSTM cell mainly comprises a cell state and three gates: the input gate, forget gate, and output gate. The input gate controls how much of the incoming information is allowed into the cell. The forget gate regulates the extent to which the previous cell state is retained or discarded. The output gate determines how much of the cell state is passed on as output. Together, these components enable LSTM networks to capture long-term dependencies within time series data. The internal structure of a basic LSTM cell is shown in Figure 4.

In Figure 4, f_t, i_t, o_t, and c_t are variables associated with the forgetting gate, input gate, output gate, and memory cell state; h_t is the output of the LSTM cell; and σ is the sigmoid activation function. The internal logic of the LSTM involves the following equations:

$$\begin{aligned} f_t &= \sigma\left(W_f x_t + U_f h_{t-1} + b_f\right) \\ i_t &= \sigma\left(W_i x_t + U_i h_{t-1} + b_i\right) \\ o_t &= \sigma\left(W_o x_t + U_o h_{t-1} + b_o\right) \\ c_t &= f_t \otimes c_{t-1} + i_t \otimes \tanh\left(W_c x_t + U_c h_{t-1} + b_c\right) \\ h_t &= o_t \otimes \tanh\left(c_{t-1}\right) \end{aligned} \tag{13}$$

where the W, U, and b with different subscripts are weight matrices and bias vectors corresponding to different gates or states.

4 | The Framework of Joint Wind–PV Power Forecasting

4.1 | Summary of the Studied Framework

Figure 5 illustrates the framework for joint power forecasting of distributed wind and PV sites, with particular consideration of the varying availability of measurements across different locations.

The framework includes three steps:

1. *Sites Classification*. The wind and PV sites are classified into three groups based on the availability of measurements. Sites with a long history of measurements (e.g., a year) are classified as reference sites (R-sites). Sites with no data records are classified as O-sites. Sites with only partial or short-term data (e.g., a month) are classified as P-sites. Newly built sites are always categorised as O-sites or P-sites.

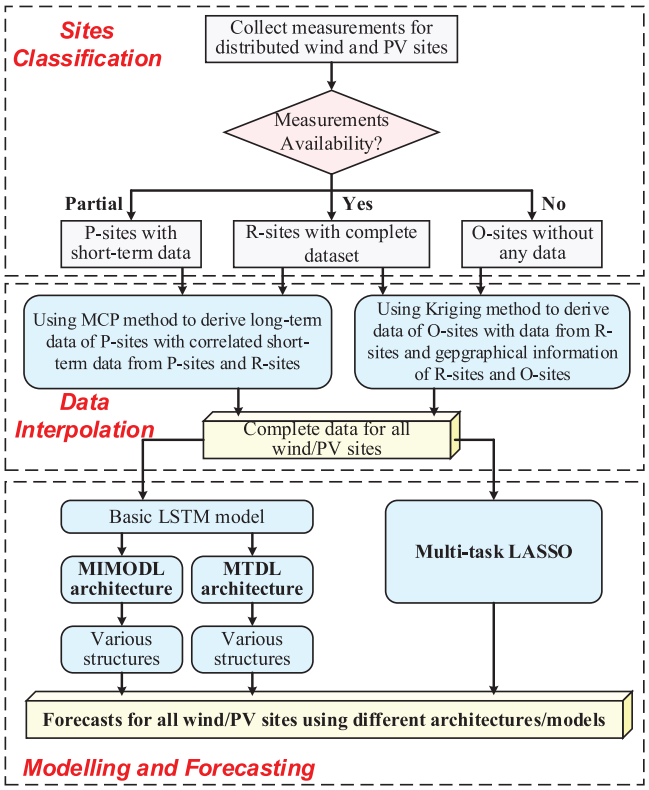


FIGURE 5 | The schematic diagram of the overall framework for joint wind–PV power forecasting.

2. *Data Interpolation*. For O-sites, the Kriging method is used to interpolate data for the times and dates of interest. This requires the geographical latitudes and longitudes of O-sites and R-sites, as well as measurements at each time point from R-sites as inputs to the Kriging model. The model will then output interpolated data for O-sites at each time point. For each P-site, the MCP model is first built using partial measurements from the P-site and synchronous measurements from one or more R-sites. The MCP model then uses long-term measurements from R-sites as inputs and provides inferred long-term data for the P-sites.
3. *Modelling and Forecasting*. The data for each site are divided into training, validation, and testing datasets. The training and validation datasets are used to train and optimise the joint power forecasting models with the three introduced forecasting architectures. The testing dataset is used to evaluate the performance of the trained models for joint power forecasts for all wind and PV sites.

4.2 | Evaluation Metrics

Two metrics are used to quantitatively evaluate and compare the performance of different forecasting models, i.e., the normalised root-mean-square error (NRMSE) and the normalised mean absolute error (NMAE).

The NRMSE is expressed as

$$e_{\mathrm{NRMSE}} = \frac{1}{C_{\mathrm{rated}}}\sqrt{\frac{1}{L}\sum_{l=1}^{L}\left(\hat{y}_l - y_l\right)^2} \times 100\% \tag{14}$$

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where L is the number of samples used for evaluation, $\hat{y}_l$ is forecast value, y_l is real measurement and C_{rated} is the rated capacity of the evaluated wind or PV site.

The NMAE is expressed as

$$e_{\text{NMAE}} = \frac{1}{C_{\text{rated}}} \frac{1}{L} \sum_{l=1}^{L} |\hat{y}_l - y_l| \times 100\% \tag{15}$$

In the following case study, to reduce the impact of uncertainty inherent in forecasting models on the comparative analysis, each experiment is conducted 10 times, and the average results are reported for each architecture, model, and forecasting horizon.

4.3 | Platform and Tools

All the architectures and models introduced in previous sections are carried out in Python 3.9.15. Specifically, the MT-LASSO is implemented with Scikit-learn 1.0.2, and the LSTM-based deep learning networks are implemented in the framework of Tensorflow and Keras 2.10.

The setting of hardware includes a 64-bit PC with 2.9 GHz Intel Dual Core CPU and 12 GB RAM. The GPU device used to support the acceleration of the deep learning networks is the NVIDIA GeForce GTX 960M.

5 | Case Study and Discussions

5.1 | Data Description

The data are generated from the open data website, renewables.ninja, which allows running simulations of the hourly power output from wind and solar power plants located anywhere in the world [34]. A total of 15 wind site locations and 15 PV site locations are randomly selected in an area of southern China. The geographical distribution of these sites is shown in Figure 6.

For each site, a MERRA-2 weather dataset is provided. The weather variable for a wind site is wind speed, and the turbine model, installed capacity and the hub height need to be specified to generate synthetic wind power data. The weather variables for a PV site are the direct irradiance, the diffuse irradiance and the temperature, and the installed capacity as well as other PV panel descriptions need to be specified to generate synthetic PV power data.

As mentioned before, either the wind sites or PV sites are randomly classified into three groups: R-sites, O-sites, and P-sites. Each group contains five sites. The classified groups are listed in Table 1. The selected R-sites for wind and PV are shown as solid triangles and solid circles, respectively, in Figure 6. The time series of each site contains 8760 data samples for the year of 2019. Though a complete dataset is generated for each site, to investigate the framework discussed in this paper, it is assumed that no data is available for the O-sites and only 744 data samples in January are available for P-sites.

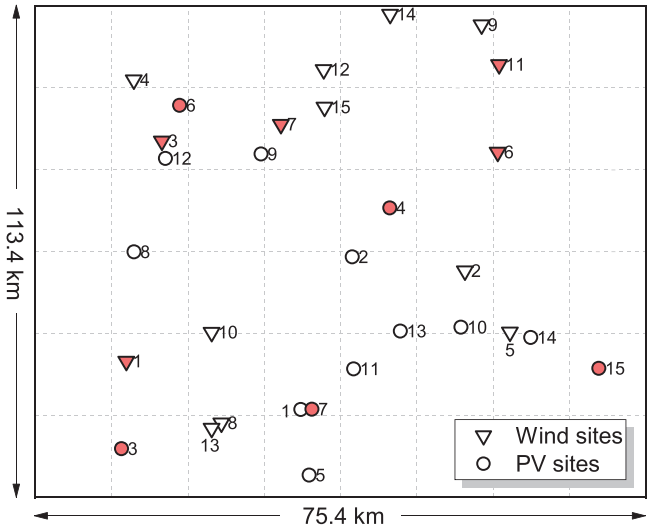


FIGURE 6 | Geographical distribution of wind and PV power generation sites.

Seventy-five per cent of the data was used for model training and hyperparameter tuning, with 80% of that portion allocated for training and 20% for tuning. The remaining 25% of the data was used for model testing and evaluation. The forecasting models are applied for six look-ahead horizons, i.e., 1–6 h. Models are built and trained separately for different horizons but with the same parameter settings. The input features for different forecasting architectures and models include the weather and power data at several continuous history lags from all 30 sites. The outputs of the forecasting models are wind and PV power forecasts for all 30 sites for a specific time horizon.

To reduce the impact of heterogeneous data on the training of forecasting models, all data are transformed into a similar range via feature-wise normalisation based on the mean and variance values of the corresponding features, which can be expressed by

$$x' = \frac{x - \bar{x}}{\sigma} \tag{16}$$

where x and x' are the original data and transformed data of a feature, and $\bar{x}$ and σ are the mean and standard variance of the feature.

TABLE 1 | Classified groups of wind and PV sites according to their data availability.

Groups	Wind power			PV power		
	R-sites	P-sites	O-sites	R-sites	P-sites	O-sites
Site index	1, 3, 6, 7, 11	2, 8, 12, 14, 15	4, 5, 9, 10, 13	3, 4, 6, 7, 15	2, 9, 11, 12, 14	1, 5, 8, 10, 13

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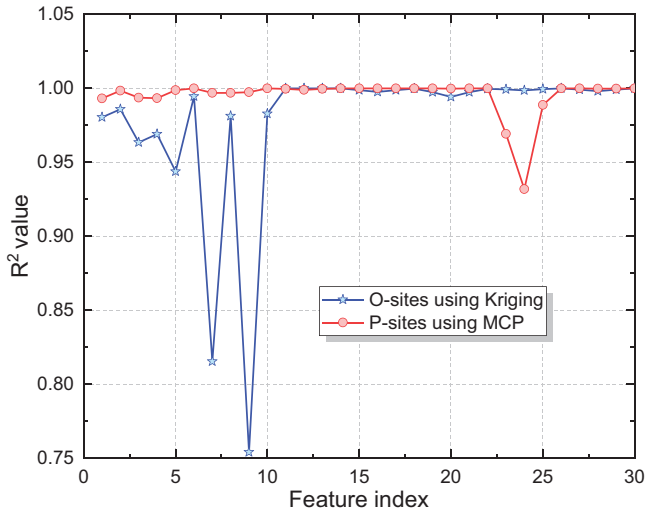


FIGURE 7 | The R^2 values computed by using the estimated and real values of features for O-sites and P-sites.

5.2 | Data Reconstruction Results

The Kriging method is used to estimate the data for O-sites using the reference data of R-sites in a feature-by-feature manner. This involves calculating the data for each feature (such as wind speed or power) of an O-site based on the Kriging using the data of the corresponding feature of all R-sites. The same process is followed to obtain the data for P-sites using the MCP method and referencing data from R-sites as well as short-term data from P-sites.

The coefficient of determination R^2 is employed to evaluate the data estimation accuracy, which is computed by using the estimated and real values of various features (e.g., wind speed, wind power) for O-sites and P-sites. R^2 is a useful measure for evaluating the goodness of fit of a linear regression model and for comparing different models. A high value of R^2 indicates that the model is a good fit for the data and that it explains a large proportion of the variation in the dependent variable. Thus, it can quantify the similarity between estimated time series and the real measurements.

R^2 values obtained when using Kriging to estimate the features of the O-sites and MCP to estimate the features of the P-sites are shown in Figure 7.

In the figure, all the R^2 values for features of P-sites using MCP are above 0.9, and most of them are close to 1.0, which shows a good estimation capability of MCP. By contrast, the performance of Kriging for O-sites is overall worse, and for two features, the R^2 is even below 0.85. However, most of the R^2 values for Kriging are above 0.95, indicating that Kriging is still a good alternative when no data is available at a target site, e.g., a newly built wind or PV site. It is also worth noting that the estimation accuracy can be substantially improved even with a limited number of short-term measurements taken at a target site.

The R^2 values are also depicted from the perspective of sites, as shown in Figure 8. The R^2 value for each site is averaged over all features of the site. It is seen that the estimation for PV O-sites is much better than wind O-sites. This is because the night portion of the time series for PV power and irradiance is constantly zero and can be estimated with 100% accuracy. On the other hand, wind speed and wind power are much more variable and challenging to estimate accurately.

When examining Figures 6 and 8 together, another important observation is that the wind or PV sites with lower estimation accuracy are generally located far away from the concentration centre of their respective groups of R-sites. For instance, in

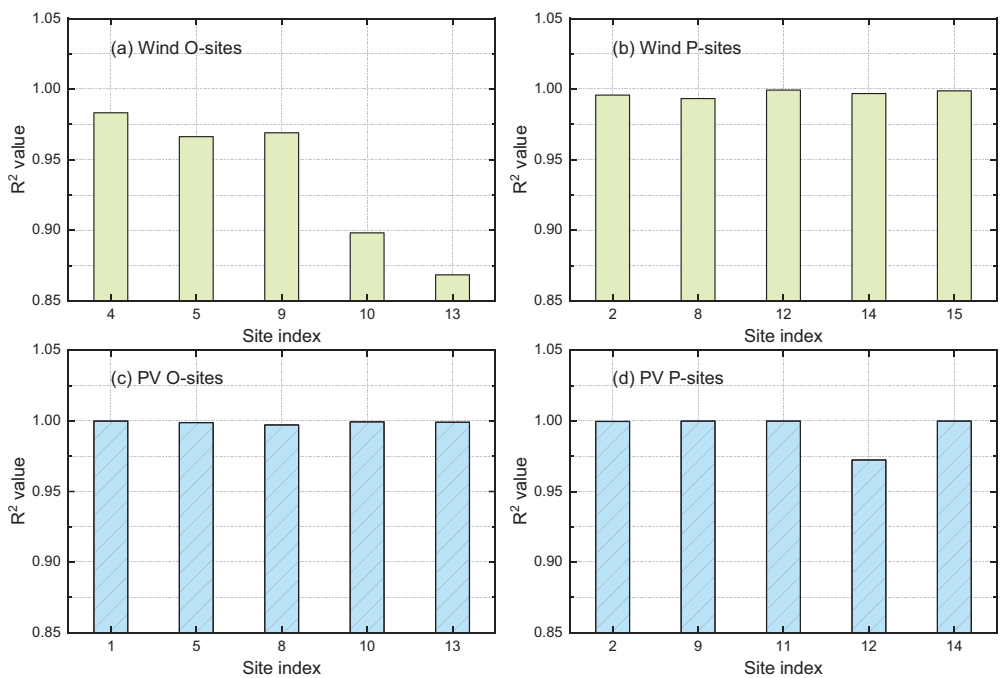


FIGURE 8 | The average R^2 values of different sites computed by using R^2 values of their features.

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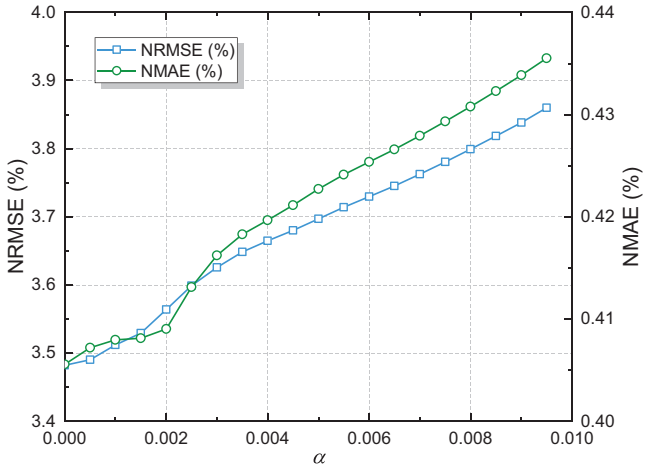


FIGURE 9 | Validation errors with varying regularisation parameters of MT-LASSO.

Figure 6, four out of five wind R-sites are located in the upper part, while the wind O-sites #10 and #13 are far away from the R-sites and have lower estimation accuracy than other O-sites. Likewise, #12 is the farthest PV P-site away from most PV R-sites located in the lower right part of the figure; it exhibits the lowest estimation accuracy. The same pattern is observed for wind P-site #8 and PV O-site #8. It is implied that when selecting or building reference sites, the area encompassed by all reference sites should sufficiently cover the sites for which data needs to be estimated so that the estimation accuracy can be ensured.

5.3 | Parameter Settings of Forecasting Models

It should be noted that the aim of this paper is not to propose the best forecasting model with optimal parameters that outperforms existing benchmarks, but rather to investigate and compare the applicability and performance of the discussed architectures and models for joint forecasting of distributed wind and PV power generation.

Based on validation data, the average NRMSE and NMAE over all sites by using MT-LASSO with different regularisation parameter α are shown in Figure 9. It is demonstrated that MT-LASSO achieves the best performance when $\alpha = 0$, which indicates that none of the 30 sites and their features are redundant for joint forecasting. It seems difficult for MT-LASSO to extract compressed shared information for all wind and PV sites by simply shrinking the model structure. Therefore, the parameter α of MT-LASSO for all forecasting time horizons is set as zero.

The two architectures, MIMODL and MTDL, will be compared for joint forecasting of wind and PV power generation. In more detail, the performance of architectures with varying numbers of layers and different combinations of layers will be discussed. The combination of layers includes the various mix of shared layers and task-specific layers for the MTDL, and the mix of LSTM layers and fully connected (FC) layers for both the MOMODL and MTDL.

The structure settings for deep learning models outlined in Table 2 will be used and compared. The main objective is to

TABLE 2 | Different structure settings for the MIMODL and MTDL.

Architecture	MIMODL	MTDL
Settings with two layers in total (**S2**)	• **S2-11:** 1 LSTM layer + 1 FC layers	• **S2-1-1:** 1 shared LSTM layer + 1 task-specific FC layer
Settings with three layers in total (**S3**)	• **S3-12:** 1 LSTM layer + 2 FC layers	• **S3-1-2:** 1 shared LSTM layer + 2 task-specific FC layers • **S3-11-1:** 1 shared LSTM layer + 1 shared FC layer + 1 task-specific FC layer
Settings with four layers in total (**S4**)	• **S4-22:** 2 LSTM layers + 2 FC layers	• **S4-2-2:** 2 shared LSTM layers + 2 task-specific FC layers • **S4-21-1:** 2 shared LSTM layers + 1 shared FC layer + 1 task-specific FC layer • **S4-1-12:** 1 shared LSTM layer + 1 task-specific LSTM layer + 2 task-specific FC layers

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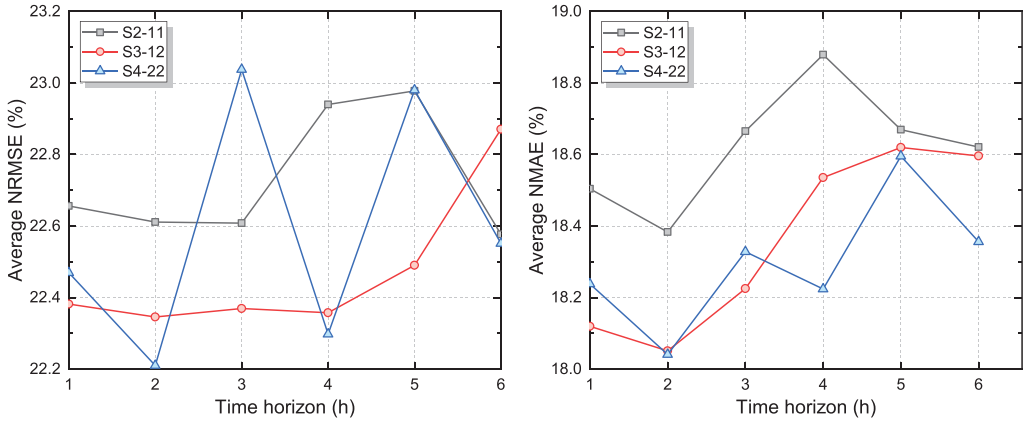


FIGURE 10 | Averaged NRMSE and NMAE of different MIMODL structures and forecasting horizons.

examine how the total number of layers and the proportion of task-specific layers will affect joint forecasting performance. It should be noted that the layers mentioned in the table do not include the input layer and the output layer. Furthermore, the number of hidden units in a shared layer or the total number of hidden units in all task-specific layers for the MTDL is equivalent to the number of hidden units in the corresponding layer for the MIMODL.

By conducting a proper grid search, the number of hidden units was set to 128 for all shared LSTM layers, 64 for task-specific LSTM layers, 256 for shared FC layers, and 64 for task-specific FC layers. Root mean-square propagation (RMSprop) was used as the optimiser. The batch size was set to 32, and the learning rate to 0.01.

In addition, two benchmark models – the persistence model [35] and the Random Forest (RF) model [36] – are employed to demonstrate the effectiveness and superiority of different architectures and models for joint forecasting. The persistence model is a naive forecasting method that assumes future power generation will be the same as the most recent observed value. RF is an ensemble learning method based on decision trees. It builds multiple trees on different subsets of the data and averages their outputs for regression tasks. After performing a grid search, the number of decision trees is set to 75 and the maximum depth to 8, while the remaining parameters of the RF follow the default settings of the RandomForestRegressor in the Scikit-learn library. It should be noted that each benchmark model is applied to one target site at a time, using only the measurements from the target site itself.

5.4 | Forecasting Results of Different Architectures

5.4.1 | Performance Analysis of MIMODL

The NRMSE and NMAE values of the MIMODL averaged over all wind and PV sites are shown in Figure 10. The average metrics are computed for different structure settings and forecasting time horizons. For comparison, the average NRMSE and NMAE values of two benchmark models are provided in Tables 3 and 4. Overall, the structures with more layers have better accuracy, and the error increases with longer time horizons. However, it is clearly seen that the average NRMSE for all structure settings and horizons falls between 22% and 23%, while the average NAME falls between 18% and 19%. By comparing with the benchmark models, this level of forecasting errors of the MIMODL is very high for ultra-short-term wind and PV power forecasting, especially for time horizons shorter than 4 h.

TABLE 3 | Average NRMSE for the two benchmark models for different time horizons.

Methods	1 h	2 h	3 h	4 h	5 h	6 h
Persistence	6.97	12.76	17.50	21.34	24.41	26.84
RF	11.38	13.07	15.09	16.86	18.14	19.06

TABLE 4 | Average NMAE for the two benchmark models for different time horizons.

Methods	1 h	2 h	3 h	4 h	5 h	6 h
Persistence	4.05	7.73	11.04	14.03	16.64	18.92
RF	5.84	7.71	9.73	11.48	12.73	13.56

Figure 11 shows some selected forecasting results using MIMODL with S4-22 structure for a 3-h ahead horizon. There is no resemblance between the curve of the forecasts and the curve of the measurements. It seems the forecast curve fluctuates around an underlying average line. It is also unexpected that the wind power forecast curve exhibits a certain period, like the PV power forecasts, and the period is about 30 h. One possible explanation for these results is that the MIMODL models are forced to learn the shared or the correlated patterns between the two energy forms with significantly different data distributions, and their structures are not allowed to learn specific patterns that are solely owned by wind or PV power. In this case, the correlation information extracted by the MIMODL models comprises average magnitude and periodicity. The periodicity is attributed to the intrinsic periodicity of PV power as well as the frequent ups and downs in wind power, which are also treated as periodicity.

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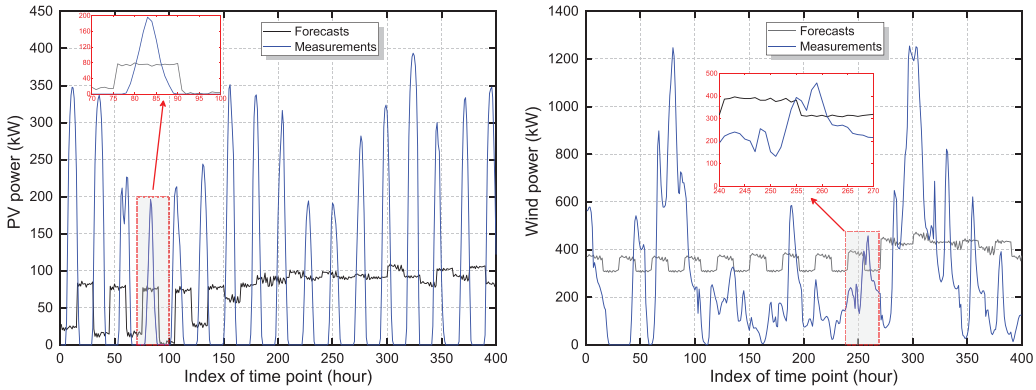


FIGURE 11 | Forecasting results using MIMODL with S4-22 structure for a 3-h ahead horizon.

TABLE 5 | Average NRMSE of MTDL for different structures and time horizons.

Horizons	S2-1-1	S3-1-2	S3-11-1	S4-2-2	S4-21-1	S4-1-12
1	9.39898	**5.33939**	7.72236	**4.94681**	6.23895	7.89969
2	13.8945	**10.6299**	13.6053	**10.6553**	13.2461	14.3413
3	17.8639	20.1182	19.2227	**17.2253**	**13.893**	18.0883
4	21.9342	21.5823	**18.566**	**18.505**	19.4033	21.4127
5	25.7591	22.6859	**20.9406**	21.5731	**20.3835**	24.7069
6	25.7518	26.1122	**24.1351**	**23.272**	25.3506	26.1677

TABLE 6 | Average NMAE of MTDL for different structures and time horizons.

Horizons	S2-1-1	S3-1-2	S3-11-1	S4-2-2	S4-21-1	S4-1-12
1	7.18363	**3.78757**	5.72759	**3.23563**	4.3246	6.23308
2	10.4085	**7.74255**	9.84936	**7.58541**	8.06037	10.8278
3	13.7249	15.686	15.1079	**13.0076**	**10.6718**	13.82
4	17.2313	17.2232	**14.932**	**13.2127**	15.2144	16.4337
5	20.192	17.7978	**17.1104**	17.137	**15.9935**	19.0857
6	19.8656	20.0801	**18.8381**	**18.1347**	20.3446	20.292

It is implied that the MIMODL architecture is not capable of extracting correlation information across the forecasting targets with distinct data distributions while retaining the specific patterns unique to each target. So, it is not a good candidate for joint wind and PV power forecasting.

5.4.2 | Performance Analysis of MTDL

The average NRMSE and NMAE of MTDL for different structure settings and forecasting time horizons are given in Tables 5 and 6. Generally, the errors increase as the forecasting horizon expands, regardless of the model structure settings. However, the performance of various structures differs significantly.

For each horizon, the error values for the two best structures are bolded in Tables 5 and 6. It can be observed that though none of the structures could achieve the best performance across all time horizons, it is still apparent that the S4-22 structure performs the best overall. It is widely acknowledged that the depth of deep learning models has a significant impact on their forecasting ability. Therefore, the total number of layers must be set appropriately.

In the context of joint forecasting discussed in this paper, the case becomes more complex. The results are not only affected by the total number of layers but also influenced by the proportion of shared layers and the task-specific layers. By comparing the structures S3-1-2 and S3-11-1 with S4-1-12, simply increasing the total number of layers does not necessarily improve the forecasting accuracy. By comparing the S4-2-2, S4-21, and S4-1-12, the forecasting accuracy is sensitive to the proportion of shared layers and task-specific layers. Overweighting the shared layers could lead to the same problem as the MIMODL, for which the abstract correlation information is emphasised while the specific patterns unique to each individual site are given less attention. In contrast, too many task-specific layers could cause overfitting of the unique patterns and lead to reduced accuracy.

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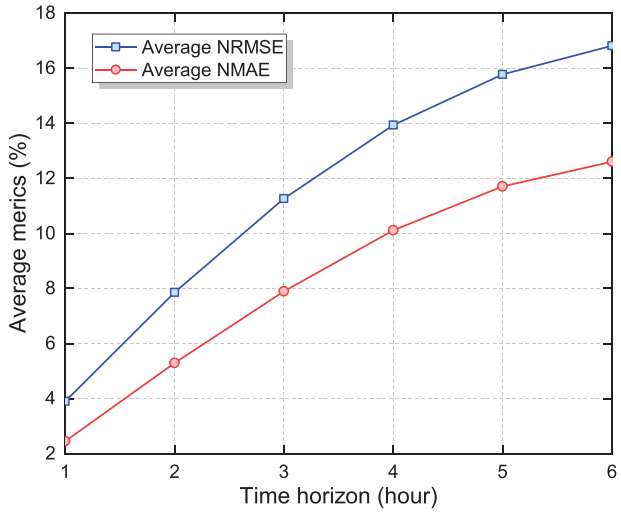


FIGURE 12 | Average NRMSE and NMAE of MT-LASSO for different time horizons.

By comparing the MTDL with the two benchmark models, the MTDL is better than the persistence model across all time horizons, and there is always a best structure that can well outperform the RF model for time horizons shorter than 5 h.

In summary, the MTDL architecture is a feasible and promising solution for joint forecasting of wind and PV power generation. However, it needs to be elaborately designed, which includes the mix of shared layers and task-specific layers, the selection of layer types such as LSTM and CNN, and various parameters such as number of hidden units and activation functions for optimal performance. Also note that the same layers and parameters are specified for different tasks in this work. However, using more specific settings for different tasks may further improve accuracy.

5.4.3 | Performance Analysis of MT-LASSO

The average NRMSE and NMAE of MT-LASSO for different time horizons are shown in Figure 12. By comparing with the two benchmark models, the MT-LASSO model consistently outperforms both the persistence and RF models across all time horizons.

In addition, the MT-LASSO performs much better than MIMODL and MTDL for all time horizons in terms of accuracy. As mentioned earlier, the MT-LASSO has a completely different multi-task learning architecture, in which model coefficients are jointly trained through a shared loss function rather than shared common coefficients. For comparison, the MIMODL has fully shared common coefficients, and the MTDL uses a combination of shared common coefficients and distinct task-specific coefficients.

These results suggest that using a shared loss function could be a highly effective approach for multi-task joint forecasting of wind and PV power generation.

6 | Conclusion

A framework of joint forecasting of distributed wind and PV power generation is studied. The goal is to build unified models to improve modelling efficiency and to leverage the correlation between sites and different energy forms. In particular, the study considers a scenario where measurements for most sites are either partially or not available. Different model architectures are compared to investigate their viability and performance for joint forecasting. Main conclusions are drawn as follows:

1. The Kriging method and MCP method are effective in estimating data for sites without any historical measurements and sites with only short-term measurements, respectively. It is shown that the estimation accuracy of the MCP method is much higher by using the short-term measurements from target sites compared with the Kriging method.
2. To ensure an accurate estimation, it is suggested that the area covered by the selected reference sites should sufficiently encompass the target sites for which data are estimated.
3. The MIMODL architecture is not feasible for joint forecasting of wind and PV power generation, as it can only extract common information shared by all sites and energy forms, which is not suitable when different sites or energy forms exhibit extremely different data distributions.
4. The MTDL architecture is applicable for joint forecasting, which allows a balance between the extracted common information for all sites and unique patterns specific for each site. However, its structures and parameters must be carefully designed and tuned to achieve a satisfied accuracy level.
5. The linear MT-LASSO performs best among the three considered architectures because it jointly trains tasks via its shared loss function while each site retains its independent set of coefficients.

Though the MTDL architecture does not achieve the best performance, it is still promising and worthy of further investigation in the future. More structures and hyperparameters can be explored. The basic forecasting model can also be replaced with other deep learning models, e.g., the convolutional neural networks. Additionally, the training loss function of the MTDL can be properly designed, drawing on the experience gained from the MT-LASSO architecture.

Author Contributions

Yongning Zhao: conceptualization, formal analysis, funding acquisition, investigation, methodology, writing – original draft. **Dezhi Li:** investigation, methodology, validation, writing – review & editing. **Ke Chen:** formal analysis, investigation, validation. **Weidong Wang:** validation, visualization. **Shiji Pan:** visualization, writing – review & editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available at www.renewables.ninja.

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